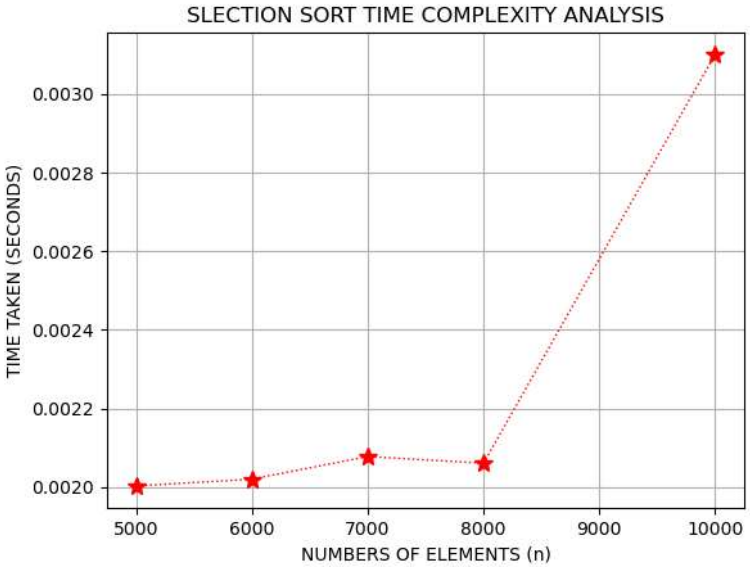
NAME - ARFHA.M.RANEBENNUR

USN - 2VX24UE012

EXPT.NO - 2

TITLE - QUICK SORT

In [1]:

```python
import random
import time
import matplotlib.pyplot as plt

def quick_sort(arr):
    if len(arr) <= 1:
        return arr
    pivot = arr[len(arr)// 2]
    left = [x for x in arr if x < pivot]
    middle = [x for x in arr if x == pivot]
    right = [x for x in arr if x > pivot]
    return quick_sort(left) + middle + quick_sort(right)

# Input:
arr = list(map(int, input("ENTER THE LIST OF ELEMENTS SEPARATED BY SPACES: ").split()))
print("INPUT ARRAY: ",arr)

sorted_arr = quick_sort(arr)
print("SORTED ARRAY: ",sorted_arr)

# MEASURE TIME TAKEN TO SORT
start_time = time.time()
sorted_arr = quick_sort(arr)
end_time = time.time()
print("TIME TAKEN TO SORT: ",end_time - start_time, "SECONDS")
print("==============================================================")

# FOR n>5000
n_values = [5000, 6000, 7000, 8000, 10000]
time_values =[]

for n in n_values:
    arr = [random.randint(1,9) for _ in range(n)] # GENERATE RANDOM ARRAY

    start_time = time.time()
    sorted_arr = quick_sort(arr)
    end_time = time.time()

    time_taken = end_time - start_time
    print("TIME TAKEN TO SORT", n, "ELEMENTS:", time_taken, "SECONDS")
    time_values.append(time_taken)
print("==============================================================")

# plotting the graph
plt.plot(n_values, time_values, color = 'red',marker = '*', markersize = 10, linewidth = 1, linestyle = 'dotted')
plt.xlabel('NUMBERS OF ELEMENTS (n)')
plt.ylabel('TIME TAKEN (SECONDS)')
plt.title('QUICK SORT TIME COMPLEXITY ANALYSIS')
plt.grid(True)
plt.show()
```

```
ENTER THE LIST OF ELEMENTS SEPARATED BY SPACES: 3 67  7 54  457 4 74 8
INPUT ARRAY:  [3, 67, 7, 54, 457, 4, 74, 8]
SORTED ARRAY:  [3, 4, 7, 8, 54, 67, 74, 457]
TIME TAKEN TO SORT:  0.0 SECONDS
==============================================================
TIME TAKEN TO SORT 5000 ELEMENTS: 0.002002716064453125 SECONDS
TIME TAKEN TO SORT 6000 ELEMENTS: 0.002020120620727539 SECONDS
TIME TAKEN TO SORT 7000 ELEMENTS: 0.0020775794982910156 SECONDS
TIME TAKEN TO SORT 8000 ELEMENTS: 0.0020606517791748047 SECONDS
TIME TAKEN TO SORT 10000 ELEMENTS: 0.003100872039794922 SECONDS
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```